Desca Affajry Rais

Mathematics Undergraduate · Researcher · Educator

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PROFESSIONAL SUMMARY

I am a final-year Mathematics undergraduate at Universitas Diponegoro with a 3.92/4.00 GPA. My primary interests are pure mathematics, especially real and complex analysis, graph theory, abstract algebra, topology, and combinatorics. I also have experience in university mathematics teaching, olympiad coaching, academic-content development, data analysis, and computational projects.

EDUCATION

Sep 2023 — Present
ONGOING

Universitas Diponegoro

B.Sc. in Mathematics · GPA 3.92 / 4.00

Chair, UKM KOMIPA UNDIP (2025); teaching and tutorial experience in Real Analysis II; active in the Mathematics Student Association; ON-MIPA delegate and mathematics competition awardee.

Jun 2020 — Mar 2023
COMPLETED

SMA Negeri 1 Sulang

Mathematics and Natural Sciences · Final Grade 86.55 / 100.00

Earned regional and national mathematics awards, represented the school in OSN Informatics, and provided peer mathematics mentoring.

RESEARCH & ACADEMIC WORKS

2026
DRAFT

Distance Spectra of Corona Graphs of Complete Graphs and Paths via Invariant Subspaces

Desca Affajry Rais · Suryoto, S.Si., M.Si.

Studies the distance spectrum of $K_n \odot P_m$ using block matrices, Kronecker products, and invariant subspaces.

2026
DRAFT

Connectivity of Int-Graphic Topology on Directed Graphs

Desca Affajry Rais · YD Sumanto · Solikhin

Characterizes connectivity of finite int-graphic Alexandroff spaces through minimal open neighborhoods and the out-neighborhood inclusion preorder.

2026
PRESENTED

Modular Total Edge Irregular Labeling of Corona Product of Cycle C_4 with Path and Null Graphs

The 16th ISNPINSA · Oral Presentation

Developed explicit labeling constructions and established exact modular total edge irregularity strength results.

2026
COMPLETED

Inequality Problem Solving Book

Desca Affajry Rais · Drs. YD Sumanto, M.Si. · First Edition · 174 pages

Proof-oriented mathematics book covering inequality theory, rigorous proofs, worked examples, and exercises. The manuscript is complete; Indonesian Intellectual Property (HKI) registration is planned.

PROFESSIONAL EXPERIENCE

Aug 2026 — Present
ONGOING

University Mathematics Tutor

Tutorkuliah.id

Tutoring in advanced undergraduate mathematics with emphasis on conceptual understanding and rigorous problem solving.

Jan 2026 — Present
ONGOING

Mathematics Olympiad Coach

KOMIPA UNDIP

University-level training in Real Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, and Combinatorics.

Apr 2026 — Jun 2026
COMPLETED

Teaching Assistant

Asosiasi Matematika Terapan UNDIP

Real Analysis tutorials, proof-oriented problem solving, and academic consultation.

Dec 2025 — Mar 2026
COMPLETED

Mathematics Item Writer

SainsIn

Authored quantitative-reasoning items, detailed solutions, instructional videos, and quality-control materials.

Nov 2025 — Jan 2026
COMPLETED

Data Analyst
Emergencyy Call
Data cleaning, SQL querying, Excel reporting, KPI dashboards, visualization, and workflow documentation.

Nov 2025 — Dec 2025
COMPLETED

Mathematics Item Writer
SNBT.ID
Authored high-difficulty Quantitative Knowledge and Mathematical Reasoning items with structured solutions.

May 2025
COMPLETED

Project-Based Virtual Intern · Data Scientist
ID/X Partners × Rakamin Academy
Built a credit-risk prediction pipeline using Logistic Regression and Random Forest.

Mar 2025
COMPLETED

Project-Based Virtual Intern · Big Data Analyst
PT Kimia Farma Tbk × Rakamin Academy
Built a BigQuery analytical table and a Looker Studio performance dashboard.

LEADERSHIP & ORGANIZATIONS

Jan 2025 — Dec 2025
COMPLETED

Chairman
Mathematics and Sciences Olympiad Club (KOMIPA UNDIP)
Led strategy, daily operations, olympiad mentoring, ON-MIPA delegation, and external representation.

Jan 2025 — Dec 2025
COMPLETED

Senior Staff, Education and Reasoning Department
Mathematics Student Association, Universitas Diponegoro
Delivered mathematics tutorials, educational content, and learning-material development.

Feb 2024 — Jan 2025
COMPLETED

Junior Staff, Education and Reasoning Department
Mathematics Student Association, Universitas Diponegoro
Coordinated Academic Delegation, mentoring for 100+ students, and received Best Junior Staff and Best Department Staff recognition.

Oct 2023 — Dec 2024
COMPLETED

Staff, Networking and Public Relations Department
KOMIPA UNDIP
Supported ON-MIPA selection, served as Mathematics Liaison Officer, and led UKM Roadshow 2024.

2020 — 2023
COMPLETED

Moderator, Mentor, and Board of Governor
Brainly
Progressed through multiple moderation and mentoring roles, including supervision of 29 apprentice moderators.

2020 — 2022
COMPLETED

Organizational Leadership
Pelajar Islam Indonesia (PII), Rembang
Served as Head of Organizational Development and later General Secretary of the Rembang Regional Board.

COMMITTEE & VOLUNTEER EXPERIENCE

Facilitator Division Coordinator
27th Undip's Mathematics Competition · Jan 2025 — Feb 2026

Curriculum Division Staff
LKMM-TD FSM 2025 · Jan — Oct 2025

Event Division Staff
Introduction, Gathering, & Adaptation Mathematics 2025 · Mar — Oct 2025

Event Division Staff
Industrial Visit Mathematics 2025 · Mar — Aug 2025

Equipment Division Staff
Math Action for Social and Education 2025 · Mar — Jun 2025

Public Relations Division Staff
Leadership Training FSM 2025 · Mar — Apr 2025

Event Division Coordinator
26th Undip's Mathematics Competition · Mar 2024 — Feb 2025

Chief Executive
UKM Roadshow 2024 · Jun — Sep 2024

Volunteer Teacher
Mathematics Clinic 2024 · Aug 2024

Equipment Division Coordinator
ON-MIPA UNDIP Selection 2024 · Oct 2023 — Feb 2024

Mathematics Liaison Officer
ON-MIPA Coaching Center 2024 · Feb 2024

SELECTED PROJECTS

Feb 2026
COMPLETED

Crowd Optimization and Smart Routing Strategy at Dufan Ancol
Operations Research · Mathematical Modeling

Applied M/M/s queueing, Little's Law, greedy optimization, and knapsack concepts; proposed 14,000 daily visitors, a <45-minute waiting-time target, and 75% maximum utilization.

Feb 2026
ONGOING

DMath Learning
Mathematics Education · EdTech

Ongoing mathematics learning platform for structured materials, guided practice, problem banks, olympiad preparation, and learning tools.

Feb 2026
COMPLETED

Bike Sharing Analytics
Python · EDA · Streamlit

Identified weekday demand peaks, weekend casual-user behavior, and environmental drivers through end-to-end exploratory analysis.

Dec 2025
COMPLETED

Financial Anomaly Detection System
Scikit-Learn · K-Means · Random Forest

Combined unsupervised pattern discovery with supervised transaction classification.

May 2025
COMPLETED

Credit Risk Prediction Analysis
SMOTE · Logistic Regression · Random Forest

Analyzed 466,285 observations and reported Random Forest performance of 99.70% accuracy, 99.94% precision, and 99.46% recall.

Mar 2025
COMPLETED

Kimia Farma Business Performance Analysis
BigQuery · SQL · Looker Studio

Built a centralized analytical table and multidimensional performance dashboard.

Mar–Apr 2025
COMPLETED

Investment Portfolio Optimization
Markowitz · Efficient Frontier · Sharpe Ratio

Analyzed portfolio allocation across KLBF.JK, HEAL.JK, and KAEF.JK using risk–return optimization.

SCHOLARSHIPS & TRAINING

Nov 2024 — Present
ONGOING

BSI Scholarship Inspirasi Batch 4
Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Feb — Jul 2026
COMPLETED

Coding Camp 2026 — Data Scientist
Coding Camp powered by DBS Foundation

Coding Camp powered by DBS Foundation

Mar — Jun 2026
COMPLETED

Microsoft Elevate Training Center
Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Jan — Mar 2026
COMPLETED

IDCamp 2025 — Intermediate Data Scientist
Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

Oct — Dec 2025
COMPLETED

IDCamp 2025 — Beginner Data Scientist
Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

CERTIFICATIONS

Indonesian Language Proficiency Test (UKBI)
Score 724 · Sangat Unggul · 21 Oct 2026 · Valid until 21 Oct 2028

UNDIP SEU English Proficiency Test
EPT 65 · TOEFL Prediction equivalent 549 · 17 Sep 2026

TOEFL ITP Prediction Test
590 / 677

AI Engineer Bootcamp

Foundations of Artificial Intelligence

Foundations of SQL

Computer Networking Fundamentals

Data Analyst — Basic Level
ITBOX · Mar 2025

Python Programming Essentials: A Beginner's Guide
Rakamin Academy · Mar 2025

Classical Cryptography for Beginners
Cyber Academy Indonesia · Jul 2024

Introduction to Information Security
Cyber Academy Indonesia · Jul 2024

Student Leadership and Management Training — FSM 2024
BEM FSM UNDIP · Apr 2024

Introduction to Data Science with Python
DQLab · Jan 2024

Python Fundamentals for Data Science
DQLab · Jan 2024

Basic Student Management Skills Training
Sep 2023

Python Programming for Beginners
CODEPOLITAN · Credential Z7R8TDD · Jul 2024 — Jul 2027

SKILLS

Mathematics & Theory

Pure Mathematics · Real Analysis · Complex Analysis · Graph Theory · Abstract Algebra · Topology · Combinatorics · Mathematical Modeling · Formal Proof Writing

Technical & Computing

Python · SQL · LaTeX · Scikit-Learn · Looker Studio · Microsoft Excel · Microsoft Fabric · Microsoft Azure

Data & Analytics

Data Analysis · Exploratory Data Analysis · Machine Learning · Data Visualization · Statistical Analysis · Data Engineering

Academic & Teaching

Research · Academic Writing · Mathematics Olympiad · Teaching · Curriculum Development · Assessment Design · Problem Solving

AWARDS & ACHIEVEMENTS

2nd Place · Calculus Competition · Universitas Halu Oleo
Oct 2026

Silver Medalist · Isaac Newton Competition 2026 · OLIMNUS × UIN Sunan Kalijaga
Jul 2026

Silver Medalist · Future Science Olympiad · POSI
Jun 2026

Bronze Medalist · Mathematical Analysis and Geometry Day XVII · Institut Teknologi Bandung
May 2026

Gold Medalist · Cendekia Natural Science Competition
Apr 2026

1st Place · ON-MIPA UNDIP Selection 2026
Apr 2026

1st Place · Indonesia Science Competition (KSI)
Mar 2026

Gold Award · KTOM March 2026
Mar 2026

1st Place · Indonesia Science Competition (InSC)
Feb 2026

4th Place · Honorable Mention 1 · MATICS Mathematics Olympiad · Hasanuddin University
Feb 2026

Silver Award · KTOM February 2026
Feb 2026

Bronze Medalist · POSI Science Competition
Jan 2026

2nd Place · Indonesian Youth Science Olympiad #6
Jan 2026

2nd Place · National Science League #2
Jan 2026

Silver Medalist · Einstein Science Competition 2026
Jan 2026

Gold Award · KTOM December 2025
Dec 2025

Special Honour Participant · International Youth Math Challenge
Dec 2025

1st Place · LOSEMA X
Nov 2025

1st Place · Olimnus Science 2025
Nov 2025

Silver Medalist · Isaac Newton Competition 2025
Jul 2025

3rd Place · ASOE 4.0
May 2025

4th Place · ON-MIPA UNDIP Selection 2025
Mar 2025

Finalist · MATHLICIOUS 2022
Nov 2022

1st Place · ALHAMBRA Mathematics Competition
Feb 2022

Bronze Medalist · Insight Arjuna Competition
Oct 2021

Gold Medalist · Dirgantara Science Competition
Oct 2021

Silver Medalist · Indonesian Youth Science Competition
Oct 2021

Bronze Medalist · Bimage Competition Season 2
Sep 2021

Silver Medalist · POSI Science Competition
Sep 2021

Top 10 · National Calculus Olympiad Nusantara
Aug 2021